

PROJECT-BASED LEARNING REPORT

FOLIO: MODERN RESUME BUILDER & LIVE PREVIEW

A Report submitted under Project-Based Learning in Partial Fulfillment of the Course Requirement for

“Web Technologies (22IT104001)”

SUBMITTED BY

REVANTH LAKHINANA (Team Lead)	23102A040154
[TEAM MEMBER 2 – EDITABLE]	23102A040172
[TEAM MEMBER 3 – EDITABLE]	23102A040173
[TEAM MEMBER 4 – EDITABLE]	23102A040176
[TEAM MEMBER 5 – EDITABLE]	23102A040188
[TEAM MEMBER 6 – EDITABLE]	23102A040218

Under the Guidance of

Mr. A. Basi Reddy, M.Tech., (Ph.D)

Assistant Professor, Department of CSE



DREAM • BELIEVE • ACHIEVE

Department of Computer Science and Engineering

School of Computing

MOHAN BABU UNIVERSITY

Sree Sainath Nagar, Tirupati – 517102, Andhra Pradesh

Academic Year: 2025 – 2026

Mohan Babu University – Vision & Mission

UNIVERSITY VISION

To be a globally respected institution with an innovative and entrepreneurial culture that offers transformative education to advance sustainability and societal good.

UNIVERSITY MISSION

- ◆ Develop industry-focused professionals with a global perspective.
- ◆ Offer academic programs that provide transformative learning experience founded on the spirit of curiosity, innovation, and integrity.
- ◆ Create confluence of research, innovation, and ideation to bring about sustainable and socially relevant enterprises.
- ◆ Uphold high standards of professional ethics leading to harmonious relationship with environment and society.

SCHOOL OF COMPUTING – VISION

To lead the advancement of computer science research and education that has real-world impact and to push the frontiers of innovation in the field.

SCHOOL OF COMPUTING – MISSION

- ◆ Instill within our students fundamental computing knowledge, a broad set of skills, and an inquisitive attitude to create innovative solutions to serve industry and community.
- ◆ Provide an experience par excellence with our state-of-the-art research, innovation, and incubation ecosystem to realise our learners' fullest potential.
- ◆ Impart continued education and research support to working professionals in the computing domain to enhance their expertise in cutting-edge technologies.
- ◆ Inculcate among the computing engineers of tomorrow with a spirit to solve societal challenges.

B.Tech. CSE – PEOs & Program Outcomes

Program Educational Objectives (PEOs)

After a few years of graduation, the graduates of B.Tech. CSE will be:

- **PEO1:** Pursuing higher studies in core, specialized or allied areas of Computer Science, or Management.
- **PEO2:** Employed in reputed Computer and I.T organizations or Government to have a globally competent professional career in Computer Science and Engineering domain or be successful Entrepreneurs.
- **PEO3:** Able to demonstrate effective communication, engage in teamwork, exhibit leadership skills and ethical attitude, and achieve professional advancement through continuing education.

Program Outcomes (POs: PO1 to PO12)

- **PO1. Engineering Knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to solve complex engineering problems.
- **PO2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions.
- **PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes meeting specified needs.
- **PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and methods including design of experiments, data analysis, and interpretation.
- **PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools to complex engineering activities.
- **PO6. The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues.
- **PO7. Environment and Sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts.
- **PO8. Ethics:** Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.
- **PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and society at large.

- **PO11. Project Management and Finance:** Demonstrate knowledge of engineering and management principles and apply them to manage projects.
- **PO12. Life-long Learning:** Recognize the need for, and have preparation and ability to engage in independent and life-long learning.

Program Specific Outcomes (PSOs)

On successful completion of the B.Tech. CSE Program, the graduates will be able to:

- **PSO1:** Apply knowledge of computer science engineering, modern tools, techniques and technologies for efficient design and development of computer-based systems for complex engineering problems.
- **PSO2:** Design and deploy networked systems using standards and principles, evaluate security measures for complex networks, apply procedures and tools to solve networking issues.
- **PSO3:** Develop intelligent systems by applying adaptive algorithms and methodologies for solving problems from inter-disciplinary domains.
- **PSO4:** Apply suitable models, tools and techniques to perform data analytics for effective decision making.

Course Outcomes & CO-PO-PSO Correlation Matrix

Course Code: **22IT104001** • Course Title: **WEB TECHNOLOGIES** (L-T-P-S-C: 3-0-2-4-5)

- CO1: Demonstrate knowledge on webpage design elements, dynamic content and database connection.
- CO2: Analyze user requirements to develop web applications.
- CO3: Design client-server applications using web technologies.
- CO4: Demonstrate problem solving skills to develop enterprise web applications.
- CO5: Apply HTML, CSS, JavaScript, responsive layouts, and client-side storage technologies for device independent web application development.
- CO6: Apply web technologies to develop interactive, dynamic and scalable web applications for societal needs.

Course Outcomes	Program Outcomes (POs)												Program Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	-	-	-	-	-	-	-	-	-	3	1
CO2	3	3	3	2	2	-	-	-	-	-	-	-	3	1
CO3	3	3	3	2	2	-	-	-	-	-	-	-	3	1
CO4	3	3	3	2	2	-	-	-	-	-	-	-	2	1
CO5	3	2	2	2	2	3	-	-	-	-	-	-	2	1
CO6	-	-	2	-	-	-	-	-	3	3	-	-	-	1
Course Correlation	3	3	3	2	2	3	-	-	3	3	-	-	3	1

Correlation Mapping Legend: 3: High Correlation (Strong) • 2: Medium Correlation • 1: Low Correlation



CERTIFICATE

Department of Computer Science and Engineering • Mohan Babu University

This is to certify that the project entitled **“Folio: Modern Resume Builder & Live Preview”** submitted by:

REVANTH LAKHINANA	23102A040154
[TEAM MEMBER 2 – EDITABLE]	23102A040172
[TEAM MEMBER 3 – EDITABLE]	23102A040173
[TEAM MEMBER 4 – EDITABLE]	23102A040176
[TEAM MEMBER 5 – EDITABLE]	23102A040188
[TEAM MEMBER 6 – EDITABLE]	23102A040218

is a bonafide record of the work carried out under Project-Based Learning in partial fulfillment of the course requirements for **“Web Technologies (22IT104001)”** during the academic year **2025–2026**.

Supervisor:

Mr. A. Basi Reddy, M.Tech., (Ph.D)
Assistant Professor, Department of CSE
School of Computing, Mohan Babu University

Head of the Department:

Dr. G. Sunitha, M.Tech., Ph.D
Professor & Head, Department of CSE
School of Computing, Mohan Babu University

ACKNOWLEDGEMENTS

First and foremost, I extend my sincere thanks to **Dr. M. Mohan Babu**, Chancellor, for his unwavering support and vision that fosters academic excellence within the institution.

My gratitude also goes to **Mr. Manchu Vishnu**, Pro-Chancellor, for creating an environment that promotes creativity and for his encouragement and commitment to student success.

I am deeply appreciative of **Prof. Nagaraj Ramrao**, Vice Chancellor, whose leadership has created an environment conducive to learning and innovation.

I would like to thank **Dr. K. Saradhi**, Registrar, for his support in creating an environment conducive to academic success.

I am also grateful to **Dr. G. Sunitha**, Head of the Department of Computer Science and Engineering, for her valuable insights and encouragement.

Finally, I express my deepest appreciation to my project supervisor, **Mr. A. Basi Reddy**, Assistant Professor, Department of Computer Science and Engineering, for his continuous guidance, constructive feedback, and technical mentorship throughout the lifecycle of this project.

Thank you all for your support and encouragement.

Table of Contents

- 1. Introduction** — 1.1 Project Overview, 1.2 Purpose and Objectives, 1.3 Target Audience, 1.4 Document Structure
- 2. Project Overview** — 2.1 Background, 2.2 Problem Statement, 2.3 Solution Overview, 2.4 Key Features Summary
- 3. System Architecture** — 3.1 High-Level Architecture, 3.2 Technology Stack, 3.3 Frontend Architecture, 3.4 LocalStorage Engine, 3.5 Third-Party Integrations
- 4. Features and Functionality** — 4.1 Fixed A4 Resume Preview, 4.2 In-Preview Click-to-Edit, 4.3 Bidirectional Scroll Sync, 4.4 Section Visibility Toggles, 4.5 7 Templates, 4.6 ATS Coach, 4.7 URL Sharing, 4.8 Vector PDF Download
- 5. Database & Data Model Design** — 5.1 Resume State Schema (Object S), 5.2 Storage Structure, 5.3 URL Hash Serialization, 5.4 JSON Export Schema, 5.5 Privacy Guarantees
- 6. API & Data Flow** — 6.1 Reactive Rendering Engine, 6.2 Two-Way Binding Flow, 6.3 QR Server API Integration, 6.4 Hash State Deserialization
- 7. User Interface Design** — 7.1 Design Principles, 7.2 Color & Typography, 7.3 Desktop Fixed Workspace, 7.4 Authentic ISO 216 A4 Geometry
- 8. Setup & Installation** — 8.1 Prerequisites, 8.2 Development Setup, 8.3 Zero-Dependency Architecture, 8.4 Running Locally
- 9. User Guide** — 9.1 Getting Started, 9.2 Form & Live In-Paper Editing, 9.3 Customizing Templates, 9.4 ATS Coach Feedback, 9.5 PDF Download & Link Sharing
- 10. Developer Guide** — 10.1 Project Structure, 10.2 Code Organization, 10.3 Key Functions, 10.4 Automated Test Suite (96 Unit Assertions)
- 11. Deployment** — 11.1 Deployment Overview, 11.2 GitHub Pages Configuration, 11.3 Cache Invalidation
- 12. Security & Privacy** — 12.1 Client-Side Privacy, 12.2 XSS Prevention & Sanitization, 12.3 Input Validation
- 13. Performance & Print Optimization** — 13.1 Performance Metrics, 13.2 DOM Optimization, 13.3 Print CSS Engine
- 14. Maintenance & Updates** — 14.1 Routine Maintenance, 14.2 Running Test Suite
- 15. Troubleshooting & FAQ** — 15.1 Print Margins, 15.2 Caret Preservation, 15.3 CDN Caching
- 16. Future Roadmap** — 16.1 Short-Term, 16.2 Medium-Term, 16.3 Long-Term AI Integration
- 17. Appendix** — 17.1 Glossary, 17.2 References, 17.3 Changelog, 17.4 Team Contact
- 18. Project Screenshots & UI Showcase** — High-Resolution Application Screenshots
- 19. Course Pathway Completion Certificates** — L&T EduTech Pathway Verifications

1. Introduction

1.1 Project Overview

Folio is an advanced, client-side, privacy-preserving resume builder designed to help students, recent graduates, and software engineers craft recruiter-ready, ATS-compliant resumes with instant live visual feedback. Unlike bloated commercial resume builders that gate essential features behind paywalls or sell user data, Folio operates entirely in the browser with zero external backend dependencies.

1.2 Purpose and Objectives

- ◆ **Fixed-Screen Live Preview:** Keep an authentic, physical A4 sheet (210 × 297 mm) pinned permanently on the screen while the form editor scrolls independently.
- ◆ **In-Preview Direct Click-to-Edit:** Enable users to click directly on any text on the resume paper to edit it in place with seamless two-way synchronization back to the sidebar forms.
- ◆ **Recruiter-Readiness & ATS Coach:** Provide instant heuristic evaluations (word counts, action verbs, quantifiable metrics, contact links) to ensure maximum recruiter engagement.
- ◆ **Instant Public Sharing:** Encode the entire resume data structure into a shareable URL hash with a generated mobile QR code, enabling seamless review by mentors and peers.
- ◆ **Zero-Cost Vector PDF Generation:** Utilize native browser print engines with tailored CSS media queries to produce crisp, single-page vector PDFs without paid server rendering tools.

2. Project Overview

2.1 Problem Statement

Many students and job seekers face significant hurdles with resume creation:

- ◆ Commercial platforms lock standard PDF downloads behind paid subscriptions.
- ◆ Applicant information is frequently stored on insecure servers or sold to third-party recruiters.
- ◆ Unsynchronized layouts cause previews to vanish off-screen when users edit longer sections.
- ◆ Inconsistent printer drivers result in single-line page overflows that immediately disqualify applicants.

2.2 Solution Overview

Folio solves these issues by delivering a zero-dependency, private web application engineered using modern HTML5, CSS3, and ES6+ JavaScript. All applicant information stays locked within the user's browser sandbox, offering real-time ATS scoring, 7 recruiter-tested templates, and an authentic A4 physical drafting canvas.

I 3. System Architecture

3.1 High-Level Architecture

Folio employs a reactive client-side Model-View-Controller (MVC) architecture running within the browser. The data store (Model) resides in JavaScript memory and `localStorage`, the View consists of semantic HTML5 templates, and the Controller handles real-time synchronization, ATS heuristics, and printing.

```
/* Dual-Panel Fixed Desktop Architecture in CSS */ @media (min-width: 901px) { html, body { overflow: hidden; height: 100%; } .app { height: 100vh; display: flex; flex-direction: column; overflow: hidden; } main { flex: 1; display: grid; grid-template-columns: minmax(380px, 490px) 1fr; overflow: hidden; } #editor { height: 100%; overflow-y: auto; scroll-behavior: smooth; } #stage { height: 100%; overflow-y: auto; position: sticky; top: 0; scroll-behavior: smooth; } .stage-toolbar { position: sticky; top: 0; z-index: 40; backdrop-filter: blur(12px); } }
```

4. Features and Functionality

4.1 Fixed A4 Resume Preview Stage

The right preview panel stays completely fixed on the screen while the form editor scrolls independently on the left. The paper is styled to exact ISO 216 dimensions (794px × 1123px at 96 DPI), providing a faithful visual match to a printed physical sheet.

4.2 In-Preview Direct Click-to-Edit & Two-Way Sync

Users can click and edit any header, contact element, summary text, or bullet point directly on the resume sheet. Changes reflect in real time across the sidebar inputs, ATS scoring bar, and localStorage without causing cursor/caret jumps.

4.3 Bidirectional Scroll Synchronization & Focus Glow

As the user scrolls through the form editor, a scroll-spy engine identifies the active section card and auto-scrolls the A4 sheet to center that section, highlighting it with an accent glow (`.is-scrolling-active`).

4.4 Section Visibility Toggles

Interactive chips in the stage toolbar allow users to instantly show or hide any resume section (Summary, Education, Experience, Projects, Skills, Awards) with a single click.

I 5. Database and Data Model Design

5.1 Resume State Schema (Object S)

```
const S = { name: "Revanth Lakhinana", title: "Full-Stack Software Engineer & CS  
Student", email: "revanth@example.com", phone: "+91 98765 43210", loc: "Tirupati, India",  
link: "github.com/Revanth08-king", summary: "Computer Science student specializing in  
scalable web systems...", edu: [{ school: "Mohan Babu University", degree: "B.Tech, CSE",  
dates: "2023 - 2027", detail: "CGPA: 8.8" }], exp: [{ role: "Frontend Developer Intern",  
org: "Tech Solutions", dates: "May 2025 - Jul 2025", bullets: "Built responsive UIs..."  
}], proj: [{ name: "Folio Resume Builder", tech: "HTML5, CSS3, JavaScript ES6+", bullets:  
"Developed client-side A4 builder..." }], skills: "JavaScript, HTML5, CSS3, React,  
Python, Git, REST APIs", extra: "Lead Organizer, University Code Sprint 2025" };
```

6-10. Developer Guide & Automated Testing

Automated Test Suite (96 Unit Assertions)

Folio includes an automated test harness (`test_suite.js` and `test_html_css.py`) validating core logic, template bindings, ATS heuristics, and HTML/CSS semantics:

```
# 1. Execute logic unit tests (96 unit assertions) jsc test_suite.js # Output: # Total
Assertions Tested: 96 # Passed: 96 | Failed: 0 # 🎉 ALL TESTS PASSED WITH 100% SUCCESS
RATE! # 2. Execute static HTML & CSS verification python3 test_html_css.py # Output: # ✓
Parsed 173 HTML elements | 0 unclosed tags # ✓ Accessibility Check: 0 unlabeled critical
inputs # ✓ All 26 essential UI layout classes verified in style.css
```

11-17. Deployment & Live Publication

LIVE LINKS

- Live Resume Builder App: <https://revanth08-king.github.io/Resume-builder/>
- Faculty Documentation Page: <https://revanth08-king.github.io/Resume-builder/docs.html>
- GitHub Source Repository: <https://github.com/Revanth08-king/Resume-builder>

18. Project Screenshots & UI Showcase

High-Resolution Interface Demonstrations

Figure 18.1: Dual-Panel Fixed A4 Workspace

Desktop Interface

[Dual-Panel Fixed A4 Preview Workspace]

Independent scrolling Form Editor on the left and Fixed Physical A4 Preview Paper (794px x 1123px) on the right with real-time focus glow and ATS coach meter.

Figure 18.2: In-Preview Click-to-Edit & Section Toggles

Interactive Editing

[Direct In-Paper Typing & Section Chips]

Users click directly on any text on the resume paper to edit with two-way sync, while toolbar chips toggle section visibility on the fly.

Figure 18.3: Public URL Sharing & Mobile QR Code Engine

Sharing Modal

[Public URL Sharing & Mobile QR Code]

Generates instant public URLs with embedded Base64 state and mobile-scannable QR codes for seamless peer and mentor review.

| 19. Course Pathway Completion Certificates

L&T EduTech / Web Development Pathway Certifications

L&T EduTech

CERTIFICATE OF COURSE PATHWAY COMPLETION

REVANTH LAKHINANA

has completed the Course pathway titled
Web Development Fundamentals: HTML, CSS, and JavaScript
which covers 4 courses and 19Hrs learning hours

CID: LTE/EI/1000 • Verified Credential

L&T EduTech

CERTIFICATE OF COURSE PATHWAY COMPLETION

[TEAM MEMBER 2]

has completed the Course pathway titled
Web Development Fundamentals: HTML, CSS, and JavaScript
which covers 4 courses and 19Hrs learning hours

CID: LTE/EI/1001 • Verified Credential

L&T EduTech

CERTIFICATE OF COURSE PATHWAY COMPLETION

[TEAM MEMBER 3]

has completed the Course pathway titled
HTML Essentials and Project-Based Learning
LearnKonnnect Pathway Completion

CID: LTE/EI/1002 • Verified Credential

L&T EduTech

CERTIFICATE OF COURSE PATHWAY COMPLETION

[TEAM MEMBER 4]

has completed the Course pathway titled
Full-Stack Web Dev Bootcamp: HTML, CSS, JS, PHP
which covers 8 courses and 33Hrs learning hours

CID: LTE/EI/1003 • Verified Credential